

SQL JOINS — Practice Assessment

Topics: Inner, Left, Right, Full, Left Anti, Right Anti, Inner Join Alternative, Left/Right Join swap equivalence, Cross Join, Self Join

Instructions

Using the schema below, first write the CREATE TABLE and INSERT statements to set up the data, then write the SQL query for each of the 10 questions. Assume MySQL/PostgreSQL syntax unless stated otherwise.

Step 1: Create the Tables

Create two tables, **Employees** and **Departments**, with the structure and sample data shown below. Some employees should have no department, and some departments should have no employees, so that NULL results are visible when testing outer/anti joins.

Table: Employees

EmpID	EmpName	DeptID	Salary
1	Arun	101	45000
2	Bhavya	102	52000
3	Chitra	101	47000
4	Dinesh	103	39000
5	Esha	NULL	41000
6	Farhan	104	55000

Table: Departments

DeptID	DeptName
101	Sales
102	HR
103	IT
105	Marketing

Note: Employee 'Esha' has no DeptID (so she won't match any department), and 'Marketing' (DeptID 105) has no employees assigned to it — these gaps are intentional so anti-join and outer-join results are visible.

Step 2: Write the Following Queries

Q1. INNER JOIN

Write a query to display EmpName, DeptName, and Salary for all employees who are assigned to a department (only matching records from both tables).

Q2. LEFT JOIN

Write a query to display all employees along with their department names, including employees who do not belong to any department.

Q3. RIGHT JOIN

Write a query to display all departments along with their employees, including departments that have no employees.

Q4. FULL JOIN

Write a query to display all employees and all departments, matched where possible, including unmatched rows from both sides. (If your database does not support FULL JOIN directly, write the UNION of LEFT JOIN and RIGHT JOIN to achieve the same result.)

Q5. LEFT ANTI JOIN

Write a query to find all employees who are NOT assigned to any department (i.e., rows present in Employees but with no match in Departments).

Q6. RIGHT ANTI JOIN

Write a query to find all departments that currently have NO employees assigned to them.

Q7. INNER JOIN — Alternative Method

Write a query that produces the same result as Q1 (an inner join), but WITHOUT using the JOIN keyword — use a WHERE clause with the comma-style join (or an equivalent alternative) instead.

Q8. LEFT JOIN written as a RIGHT JOIN (and vice versa)

Take your query from Q2 (LEFT JOIN of Employees with Departments) and rewrite it as an equivalent RIGHT JOIN by swapping the table order. Then take your query from Q3 (RIGHT JOIN) and rewrite it as an equivalent LEFT JOIN by swapping the table order.

Q9. CROSS JOIN

Write a query to generate the Cartesian product of Employees and Departments, showing every possible combination of EmpName and DeptName. State how many rows the result will contain, based on the number of rows in each table.

Q10. SELF JOIN

Add a new column ManagerID to the Employees table (an EmpID referring to another employee in the same table). Then write a self-join query to display each employee's name alongside their manager's name.

Bonus (Optional)

In one or two sentences, explain the difference between a LEFT JOIN and a LEFT ANTI JOIN, and why an anti join can be thought of as 'LEFT JOIN minus INNER JOIN'.